
Perceiving Event Structure in Brief Actions

Zekun SUN & Samuel D. McDOUGLE
Yale University

Demo page:

<https://zk.actlabresearch.org/segmentation>

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Zekun Sun (*zekun.sun@yale.edu*)

Samuel D. McDougale (*samuel.mcdougale@yale.edu*)

Department of Psychology

Yale University

100 College St.

New Haven, CT 06510

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Abstract

Event segmentation is a fundamental component of human perception and cognition. The field of event cognition studies how people decide where distinct events occur in incoming sensory data, how these “event boundaries” alter decision-making and memory processes, how events reveal themselves in neural activity, and how events may be represented within perception itself. The latter point is critical — the representation of events in the first place is filtered through perception. But what counts as a minimal event that is perceptible to humans? And to what extent is the perceptual representation of minimal events driven by physical properties within stimuli (e.g., sudden changes of a tennis ball’s direction when one player strikes it) versus the semantic structure of events (e.g., “step one” versus “step two” of a tennis serve)? Here, across seven preregistered experiments, we explore the perceptual representation of event structure *within* single brief actions, and dissociate the roles of visual features and semantic structures in the perceptual segmentation of minimal events. First, participants produced boundary labels by segmenting videos of brief physical actions (e.g., kicking a ball). Then, separate groups of observers were asked to visually detect subtle disruptions in the video clips, unaware that the disruptions systematically occurred at boundary versus non-boundary timepoints. The results consistently showed an interfering effect of event boundaries on the detection of disruptions, suggesting a spontaneous perceptual representation of action structure even in very brief single actions. Moreover, boundary effects were strongest when stimuli were presented in recognizable forms versus distorted forms that only preserved lower-level features. Thus, automatic and rapid perceptual segmentation of single actions that only last several seconds may be driven by both sensory cues and our internal models of the world.

Keywords: event segmentation; action perception; schematic structure; spatiotemporal dynamics; motion

1. Introduction

The human mind tends to represent continuous experiences as discrete events, imposing “event boundaries” on incoming streams of sensory data. This phenomenon, known as event segmentation, not only shapes our perceptual experience but also underlies many cognitive processes (Zacks et al., 2007). Without it, we couldn’t easily follow the arc of a story (Kumar et al., 2023; Ünal et al., 2021), recognize recurring phrases in music (Bregman, 1994), make useful predictions (Antony et al., 2021; Zacks et al., 2007), plan complex sequences of actions (Richmond & Zacks, 2017), or reason effectively about cause and effect (Shin & DuBrow, 2021; Zacks & Tversky, 2001). Moreover, developmental research suggests that the perceptual ability to parse dynamic, continuous actions into discrete units is a fundamental cognitive tool that emerges as early as 6–11 months of age (D. A. Baldwin et al., 2001; Hespos et al., 2009; Saylor et al., 2007).

While event segmentation occurs across modalities, it plays a particularly central role in vision — our primary channel for learning about the world. We naturally interpret streams of visual input as meaningful, bounded episodes. In everyday life, a great amount of visual input involves observing other people’s actions — pouring a drink, opening a door, or tying a shoelace. Research shows that observers can reliably identify the breakpoints, or boundaries, while observing everyday activities (e.g., making a bed, washing dishes, pitching a tent), suggesting that ongoing behavior is represented as a sequence of discrete actions with defined beginnings and endings (Buchsbaum et al., 2015; Hard et al., 2011; Newton, 1973; Newton et al., 1977; Zacks et al., 2001). Event segmentation can also occur at multiple levels of granularity (Baldassano et al., 2017; Kurby & Zacks, 2008; Zacks et al., 2007, 2009): For example, a long jump can be further segmented into sub-actions such as “running,” “jumping,” and “standing.” (Figure 1A). But what are the lower limits of single action events, or “minimal events”? Can perceptual event segmentation also occur even *within* single action events?

The present study investigates perceptual event structure within single, brief actions (e.g., a jump, a step, a golf swing; see Figure 1B), with the goal of understanding event segmentation at this so-called “atomic” level. We approach this by examining the structure of action events at, to our knowledge, the shortest timescales studied to date. In addition to investigating event cognition in these minimal events, we further probe what sources of information drive the representation of action structure in visual perception.

1.1. Event structure within brief “atomic” actions

Although there is no clear definition, empirical work has shed light on the most basic units in event structures that unfold over time (see (Yates et al., 2023) for a theoretical discussion on ‘what counts an event’). For example, seminal work in event perception described the types of brief, single basic actions that constitute the “atomic components of events.” (Zacks & Tversky, 2001) In studies where observers were asked to identify the most fine-grained events in an ongoing sequence of actions, the minimal segmentable events appeared to be single actions with median lengths of 8 – 15s; these included actions like grabbing a cup, closing a door, or opening a box (Zacks et al., 2001, 2009). Similar work using short clips of these so-called atomic actions (e.g., drink, poke, blow) provided further evidence for boundary perception, not within, but *across* atomic actions (D. Baldwin et al., 2008; Buchsbaum et al., 2015; Hard et al., 2019). More recently, it has been proposed that a key feature of atomic actions is being “bounded”, a concept analogous to objects in the spatial domain. That is, bounded events are defined as actions that are internally structured and have a well-defined endpoint, such as “stacking a deck of cards” or “folding a handkerchief” (in contrast to an unbounded action, such as “waving a handkerchief”) (Lee et al., 2024).

Are such atomic actions the minimal, indivisible structure in event perception? Here we suggest that event structure can also be perceived *within* single, brief actions. To illustrate, consider simple motor skills, such as a golf swing, a tennis serve, or a swimming stroke. Although these actions can be brief (in the range of one to several seconds) and typically involve continuous, smooth body movements, they are often learned by breaking them into even smaller subroutines, e.g., a backswing and a downswing in golf (Figure 1B). Do such “steps” within a brief action act as separable events in visual processing?

Research on motor skills and sports points to a defined event structure within even the briefest single actions. Some early work suggested that the mind encodes continuous body movements as a sequence of action units defined by a preparatory-completing structure — that is, a relatively stable motion followed by an unstable motion, such as a few quick steps followed by an arabesque in ballet. Evidence for this schematic representation was revealed using visual recognition tasks: After viewing movies of a ballet dancer, observers were more accurate in identifying preparatory-completing units than completing-preparatory units (Lasher, 1981). In a similar vein, researchers have shown that elite athletes can predict the outcome of others’ actions based on early preparatory kinematic cues. For example, compared with novices, skilled basketball and volleyball players are more accurate anticipating the fate of the ball before it has even left the player’s hand, relying on the perception

of body kinematics during the preparatory phase of the action, prior to a putative event boundary (Smith, 2016; Urgesi et al., 2010, 2012). Physiological and neural data further point to the special status of preparatory phases in simple actions (Cohn et al., 2017; Urgesi et al., 2010).

Theoretical work in semantic analysis has likewise proposed a “preparation-head-coda” hierarchical structure to explain how ordinary actions are encoded and stored in memory. For example, the action sequence “making coffee” can be segmented into a preparation phase – “get out coffee,” a head phase – “measure coffee,” and a coda phase – “put away coffee”, and, critically, each phase can be further segmented into minor prep-head-coda structure (Jackendoff, 2007). These ideas and findings suggest that a multi-phase causal structure may underlie our perception of even the briefest bounded actions. However, it remains unclear whether such granular action structure is automatically represented in visual processing. In the present work, we use psychophysical methods to examine event structure across a wide range of atomic actions.

1.2. Sensory cues and semantic content in perceptual segmentation

If atomic actions are indeed subdivided into discrete events, what information does the mind use to segment them? Both theoretical and empirical work suggests that event segmentation relies on a combination of sensory cues and semantic knowledge (Buchsbaum et al., 2015; Newton et al., 1977; Zacks et al., 2007). Sensory cues for separable events are often salient, such as someone changing position in the kitchen as they move between the coffee maker and the sink. Motion cues are especially important, driving the perception of both fine-grained action events (e.g., 8 – 12 seconds, Zacks et al., 2009) and abstract events (e.g., simple animations with moving shapes, Pomp et al., 2024; Zacks, 2004). In one study, observers reported similar event boundaries in upright and inverted displays of action sequences, with these boundaries predicted by kinematic variables such as changes in direction, velocity, and acceleration of the relevant effectors (Hemeriksen & Thill, 2011).

Meanwhile, semantic event structure, e.g., “step one” versus “step two” of a tennis serve, could in theory be represented in perception, consistent with evidence that perception encodes abstract, structured, and sophisticated representations in other domains (Chen & Scholl, 2016; Cohn et al., 2017; Dasser et al., 1989; Hafri et al., 2018; H. Ji & Scholl, 2024; Loucks & Pechey, 2016; Scholl & Tremoulet, 2000; Strickland & Scholl, 2015). Although both sources of information — sensory cues and semantic knowledge — can influence event segmentation, it remains unclear how they may function separately or jointly in perceptual segmentation at the shortest time scales. A key challenge is that these two sources of information are often

111 correlated: semantically meaningful event boundaries frequently coincide with salient
112 visual feature changes. This is evident in prior work that focused on perceptual
113 effects driven by mental representation of event boundaries (Huff et al., 2012; Y. Ji
114 & Papafragou, 2022; Yates et al., 2024). For example, in a soccer game, an event
115 boundary could correspond to one player passing the ball to another — or simply to
116 a sudden change in motion cues in the scene (Huff et al., 2012). Dissociating these
117 two sources requires removing semantic content from visual streams while preserving
118 key physical stimulus properties; building on previous efforts (Pomp et al., 2024;
119 Zacks et al., 2009), our work here introduces simple new methods to do so.

120 **1.3. The present study: Event structure within atomic actions**

121 Here we ask whether event structure within brief atomic actions is represented in
122 visual perception, and to what extent spatiotemporal cues versus semantic content
123 contribute to visual segmentation. We used simple animations of single actions origi-
124 nally created using motion capture technology. These action clips are not only much
125 shorter than the stimuli used in previous studies of event segmentation, but they also
126 occur in a stripped-down context that only shows a single agent’s bodily motion. The
127 simplicity of these stimuli allowed us to manipulate semantic content while preserv-
128 ing most lower-level visual features, such that semantic content and motion cues
129 could be decorrelated. To foreshadow the key results, across seven experiments we
130 find that perceptual boundary effects consistently emerged even within brief atomic
131 actions, and that such effects were significantly stronger when the semantic structure
132 of the actions was preserved. These findings suggest that the timescale of minimal
133 events may be briefer than previously believed, and that both semantic knowledge
134 and visual cues contribute to automatic perceptual segmentation even at these brief
135 timescales.

136 **1.4. Open Science Practice**

137 This research received approval from Yale University’s local ethics board. For
138 all experiments, we pre-registered the sample size, experimental design, and statis-
139 tical analyses. We report all of the preregistered analyses; however, we also report
140 several exploratory analyses suggested by reviewers. Where relevant, we indicate
141 below which results reflect preregistered analyses and which are exploratory. The
142 data, experiment code, stimuli, and experiment pre-registrations for all studies are
143 available at: <https://osf.io/j85fa/>.

144 Demonstrations of all experiments and the full set of dynamic stimuli can be
145 found at: <https://zk.actlabresearch.org/segmentation>.

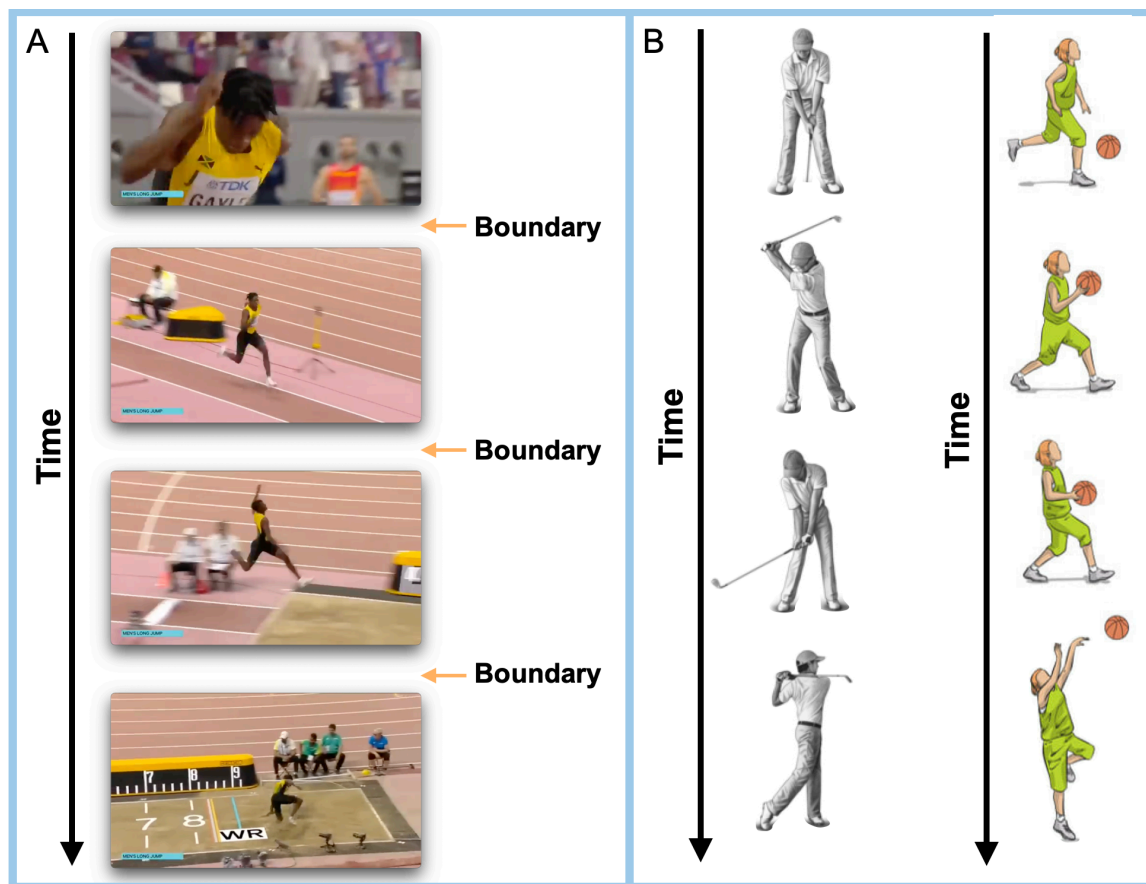


Figure 1: (A) Mental representations of event boundaries emerge from ongoing perceptual processing of continuous sensory data. As shown here, a long jump video could be segmented at the shot cut and at the transition between run-to-jump and jump-to-stand — timepoints that coincide with abrupt changes in visual features and semantic information. How short can actions be to still retain event structure in perception? (B) Here, we explore these possibilities for the case of single, brief “atomic” actions, such as a golf swing or a basketball shot. We ask whether and how the discrete ‘steps’ within a single brief action are represented in visual perception.

146 2. Experiment 1: Explicit segmentation

147 Experiment 1 aimed to identify the most characteristic boundary frame of the
148 brief “atomic” actions in our stimulus set. We asked observers to deliberately seg-
149 ment a variety of action videos into two semantically distinct units. The simple
150 MoCap animations used here appear as short, fluent and natural actions, stripped
151 of contextual cues about scenes, agents, or objects. Given that the video clips used
152 here could be very short, observers chose event boundaries offline, instead of during
153 the video clip, such that our estimates of boundaries could be more accurate. We
154 expected participants’ boundary judgments to reflect how they explicitly represent
155 the discrete steps of observed actions in a top-down manner (e.g., “step one” and
156 “step two” of a tennis serve).

157 2.1. Method

158 2.1.1. Participants

159 As stated in our pre-registration, we recruited 100 participants from the online
160 platform Prolific (<https://www.prolific.co/>). Participants were pre-screened for a
161 minimum approval rate of 99%, at least 50 prior submissions, normal or corrected-
162 to-normal vision, fluency in English, and U.S. residence.

163 2.1.2. Stimuli

164 We compiled 20 animations depicting short, natural actions, spanning sports,
165 simple exercises, and everyday tasks, from the CMU *MoCap* database([http://mocap.
166 cs.cmu.edu/](http://mocap.cs.cmu.edu/)). These videos involve an unidentified and skeletonized human figure
167 performing a single, well-defined action (e.g., making a golf swing, picking up an
168 object, taking a step aside) on a black background (Figure 2). Such simple, short
169 actions have a salient basic structure, and are thus often referred to as “bounded
170 events,” which reflect events that lead to a salient start and endpoint that is naturally
171 achieved unless there is a surprising interruption (Comrie, 1976; Y. Ji & Papafragou,
172 2022; Mittwoch, 2013). The approximate duration of the 20 actions ranged from 1.0
173 s to 3.9 s (Mean = 2.2 s, SD = 0.7 s).

174 The action videos (200×200 pixels) were displayed in participants’ web browser.
175 The workspace covered 500×500 pixels with a black background. Because of the
176 nature of online studies, we could not know the exact viewing distance, screen size,
177 and luminance (etc.) of these stimuli as they appeared to participants. However,
178 any distortions introduced by a given participant’s viewing distance or particular
179 monitor settings would have been equated across all stimuli and conditions.

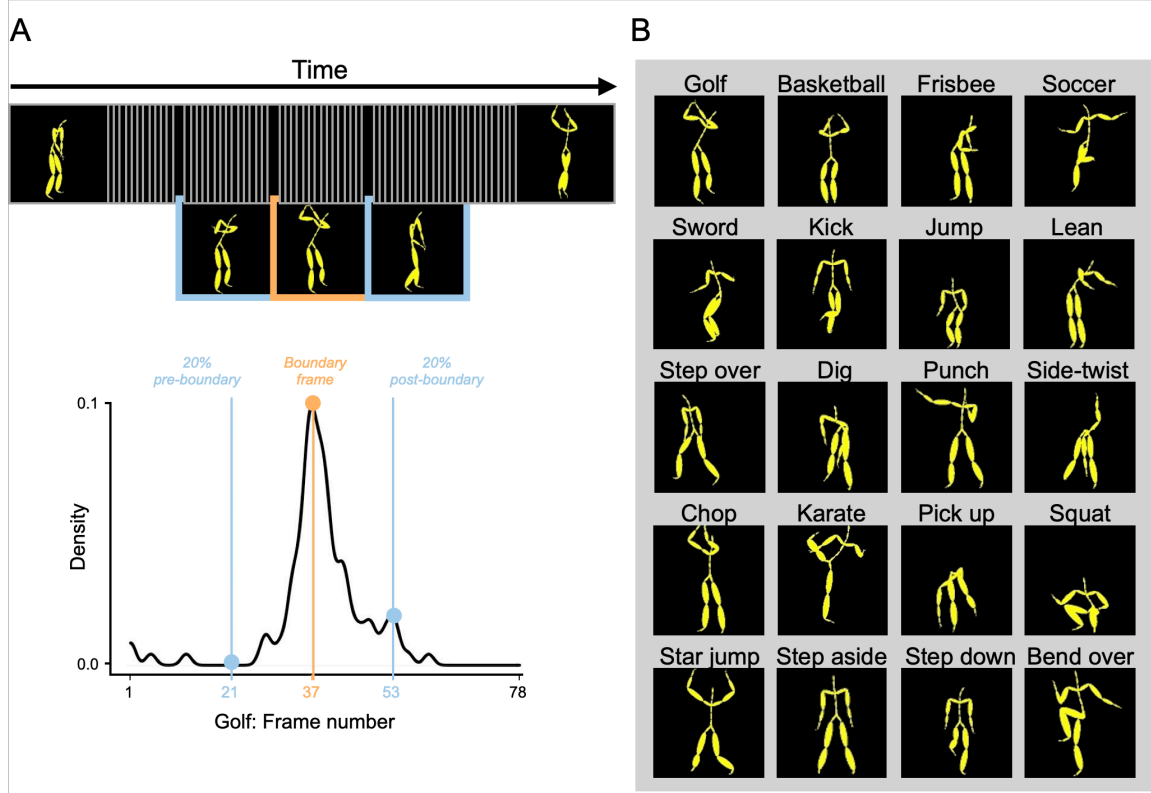


Figure 2: (A) The method used to define the most characteristic event boundary for an action. Taking the golf swing video as an example, all event boundary responses from participants were combined and smoothed with a Gaussian kernel. The frame closest to the stimulus' global peak of the event boundary choices was identified as the boundary frame, and 20% frames of the video clip before the boundary as the pre-boundary frame and 20% after the boundary as the post-boundary frame. (B) The 20 actions used in this study. Each image represents the boundary frame of each action video, as determined by our analysis method in Experiment 1.

2.1.3. Procedure

Each trial started with a “ready” cue appearing at the center of the workspace, which reminded participants to observe the full video of an action. After viewing the full video, they were given a slider they could toggle with their mouse, which allowed them to iterate through all the frames of the previously viewed action. Their job was to move the slider in order to find the most appropriate frame that divides each action into two units, such that each unit is “meaningful.” The width of the slider was determined by the number of frames of the current action video, such that the amount of mouse movement required for seeing each frame was equal across all actions. The starting position of the slider was randomized on each trial. Each participant made these boundary judgments for each of the total 20 unique actions. No time pressure was applied to participants’ responses. The serial order of actions was randomized across participants.

2.2. Results and discussion

Two participants were excluded for failing to submit a complete data set, leaving 98 participants with analyzable data. For each action, participants’ choices of the boundary were combined and smoothed with a Gaussian kernel (bandwidth = 1 frame), which gave us the density of choices for each frame of the action video. The boundary of an action was defined as the frame corresponding to the highest peak of the fitted density function (see an example in Figure 2A). Using this procedure, we obtained a single event boundary frame for all 20 actions (Figure 2B). Observers showed high agreement in selecting boundaries: On average, 40.3% observers chose either the group-level peak frame or its neighboring frames as an action’s event boundary.

Boundary selections were not limited to the videos’ middle frames: The earliest boundary was 29.4% from the beginning across all videos, and the latest was 61.3%. Indeed, as predicted given our instructions, observers typically selected boundary frames that best described how to perform an action in discrete steps, e.g., kicking a soccer ball is divided into planting the supporting foot and then swinging the kicking foot, performing a simple jump requires bending the knees and then extending upward, etc. Figure 2B depicts each boundary frame. As can be seen, the segmentation in some actions reflected the action schema that has been proposed in previous work, such as a preparing-completing structure in Lasher (1981), or prep-head-coda structure in Jackendoff (2007). For example, “shooting a basketball” was divided into a preparation phase and an execution phase, “picking up” appeared to have a “reaching” preparation phase and a “picking-up” phase, and “karate kick” appeared to reflect a head-coda structure (kicking, then withdrawing the foot). But other

cases did not fit into obvious action schema, such as the exercise skills stimuli and the simple stepping stimuli. We note that the current work does not focus on any particular type of action schema, but aims to encompass a broad range of actions and potential event structures.

3. Experiment 2: Temporal change detection

Experiment 2 explored whether the action boundary was also spontaneously represented in visual perception, rather than just reflecting an explicit decision about where the boundary belonged. Previous empirical evidence has shown that perceptual processing of event transitions (i.e., boundaries) transiently impairs one’s ability to detect subtle changes in continuous perceptual input (Huff et al., 2012; Y. Ji & Papafragou, 2022; Repp, 1992, 1998; Reynolds et al., 2007; Yates et al., 2024; Zacks et al., 2007). If action boundaries are automatically encoded during visual processing, then observers’ performance should be influenced by action structure even when the task does not demand attention to it. Here, we instructed observers to detect a transient slowdown in action videos, and tested whether their perceptual sensitivity differs when the slowdown occurs at boundary versus non-boundary frames.

3.1. Method

3.1.1. Participants

In line with our pre-registration, 20 participants were recruited through Prolific. A smaller pilot suggested that this sample would have power above 95% to reveal perceptual boundary effects.

3.1.2. Stimuli and procedure

Participants were instructed to detect transient slowdowns in the animated actions used in Experiment 1. On each trial, observers first watched the full video of an action, followed by a mask image (a yellow random noise pattern) for 900 ms, and then the video was played a second time. The second play was either identical to the first play or contained a transient slowdown (a 60- or 90-ms pause on two-thirds of trials). Observers pressed either F or J on their keyboard to indicate whether the second play did or did not contain a brief pause compared to the first play, with no time pressure to respond (Figure 3A). Critically, the pause occurred at one of three specific frames determined by the results of Experiment 1: the pre-boundary frame (20% of frames earlier than the boundary), the boundary frame, or the post-boundary frame (20% of frames later than the boundary). The three frame

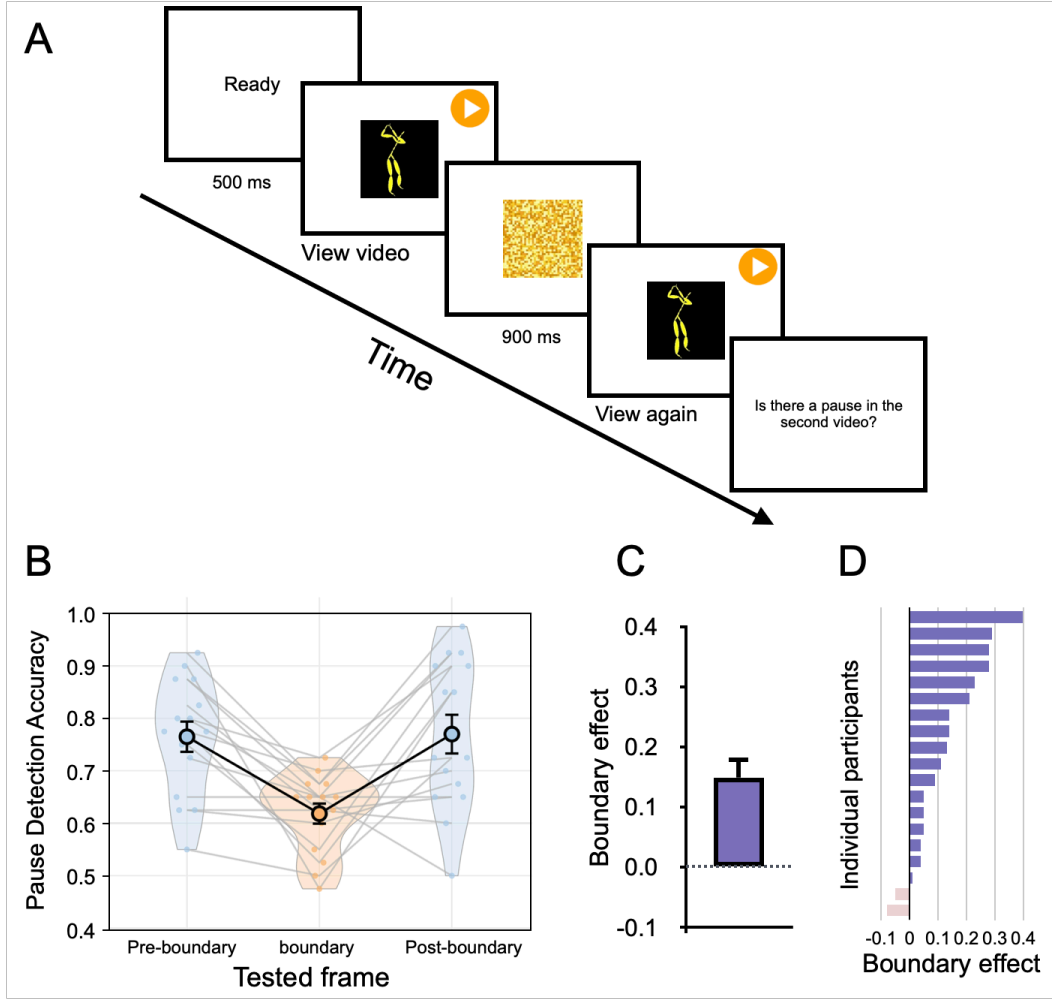


Figure 3: Trial structure and results of Experiment 2. (A) In each trial, observers watched each action video twice. During the second viewing, a brief pause could happen at one of the three tested frames (pre-boundary, boundary, or post-boundary frame). Observers judged whether there was a pause or not in the second play. (B) Compared to the pre-boundary and post-boundary frames, observers were less likely to identify the pause occurred at the boundaries. (C) Mean accuracy reduced by 14.9% at the boundary of actions across participants. (D) A strong majority of participants were less sensitive to the disruptions at the boundary of actions. The boundary effect was calculated by subtracting the mean accuracy in the boundary condition from that in the non-boundary condition, where non-boundary accuracy was averaged across pre- and post-boundary trials. Error bars = 1 s.e.m.

conditions were crossed with the 2 pause time conditions, yielding 3 (pre-, on-, post-boundary) $\times$ 2 (60, 90 ms), or 6 unique trial types. All 20 actions appeared as each of these trial types, for $20 \times 6 = 120$ trials. Additionally, each action was shown three times without a pause, adding 60 control trials. Trial order was randomized across participants. Participants were also given 3 practice trials at the onset of the task (using an action video that did not appear elsewhere in the experiment).

3.2. *Results and discussion*

Four participants were excluded for low accuracy ($< 60\%$; i.e., not significantly above chance according to a binomial test with $\alpha = .05$) and 1 for failing to provide complete data, leaving 15 participants with analyzable data. Across all participants, the mean accuracy was 75.3%. We only analyzed the trials with a pause (60 ms and 90 ms).

As shown in Figure 3B, observers were less accurate in detecting pauses at boundary frames relative to non-boundary frames (averaging pre- and post-boundary frames), $t(14) = 4.52$, $p = .00049$, Cohen’s $d = 1.17$, 95% $CI_{effect} = .15[.084, .21]$ ¹. Thirteen out of 15 participants (87%) showed lower visual sensitivity to subtle disruptions at action boundaries. Observers were more accurate in detecting longer pauses, and their accuracy was reduced at boundary frames in both 60- and 90-ms conditions (60-ms: 57.7% for boundary vs. 70.7% for nonboundary; 90-ms: 66% for boundary vs. 83% for nonboundary).

These results suggest that the transition of events within single atomic actions impedes observers’ ability to detect the temporal disruptions in visual input. Note that the cover task here did not demand observers to intentionally pay attention to event structure or boundaries of video clips, as they only had to judge whether two videos were the same or different. Thus, the boundary effects here did not reflect observers’ explicit knowledge (as in Experiment 1) but rather a spontaneous representation of action structure in visual perception. Considering the simple nature of the stimuli (e.g., a single continuous scene with no objects, perspective changes, or dramatic transitions), two possible sources of information could independently or jointly drive such perceptual effects: a) subtle spatiotemporal changes and motion cues in these dynamic scenes, and b) the semantic structure of the actions.

¹The difference scores between boundary and non-boundary conditions were normally distributed, without violating the normality assumption of our pre-registered t-tests (Shapiro-Wilk normality test: $W = 0.97$, $p\text{-value} = 0.90$). (We note that for each t-test reported throughout, we confirmed the data were normally distributed (all P s $> .1$).)

4. Experiment 3: Spatial change detection

The previous experiment addressed visual segmentation using dynamic stimuli (i.e., brief videos of actions). However, evidence exists showing that the mind can also form rich event representations from a series of discrete images, where event segmentation cannot rely on motion or dynamic signals (D. Baldwin et al., 2008; Cohn, Holcomb, et al., 2012; Ezzyat & Clements, 2024; Zheng et al., 2020). For example, people segment narratives while reading the static images of a comic book in a certain order (Cohn, Paczynski, et al., 2012). Here, we tested if perceptual boundary effects similar to those reported in Experiment 2 can emerge from such a scenario, where no motion signals can be used. In this experiment, we asked participants to detect subtle changes of spatial features instead of temporal disruptions as in the previous experiment. We predicted that the detection of spatial change would also be weakened at boundary frames if the transition between action steps is indeed represented in perception.

4.1. Method

4.1.1. Participants

As stated in our pre-registration, 20 participants were recruited through Prolific.

4.1.2. Stimuli and procedure

This experiment asked participants to view a sequence of discrete images taken from the original action videos (maintaining the temporal order) and then judge whether the last image shown changed from the previous image. The images were selected from the frames of the action videos used in Experiments 1-2, selecting one out of every three frames to create the sequence. We sequentially presented one frame from every three frames before the designated “key” frame, which was either the boundary, pre-boundary, or post-boundary frame obtained in Experiment 1 (e.g., in Figure 4A, the orange bordered frame is the boundary frame of the action “tossing a frisbee”). Following the key frame was a testing frame, which was either the same as the key frame, or 2 frames forward in time from the key frame. Thus, from the viewpoint of the observer, on each trial, a series of images was sequentially displayed (each for 700 ms, with a 100 ms ISI), and they simply needed to determine whether the last frame changed from its previous frame or not (Figure 4A). Since observers were not informed about how many images would be shown on each trial, they had to monitor every change of the images.

On each trial, participants pressed a key to observe the sequence of images depicting each action. After the last image was displayed, participants pressed either F

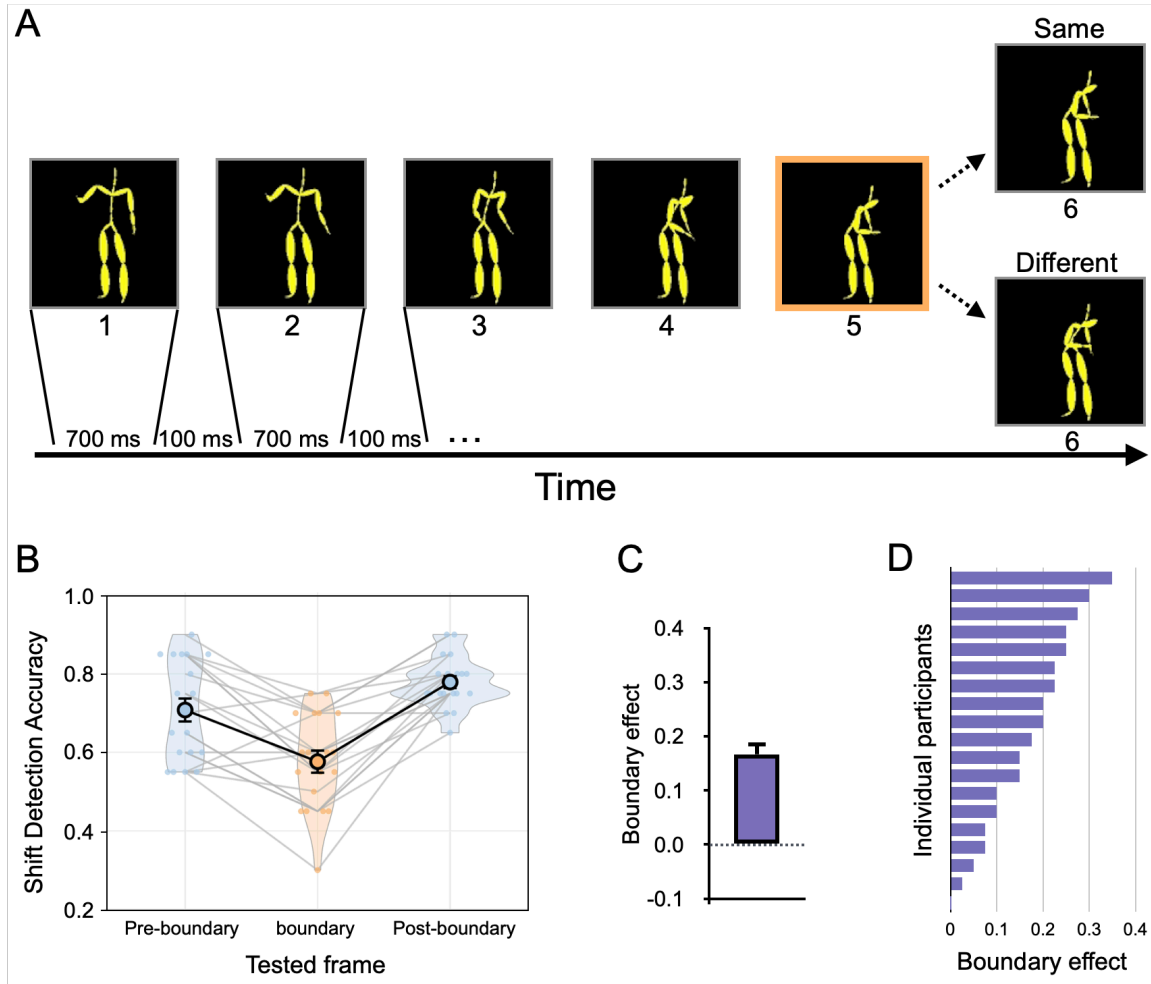


Figure 4: Trial structure and results of Experiment 3. (A) In each trial, participants observed a sequence of images (i.e., individual frames of each action video) turning on and off, with 700 ms duration and a 100 ms ISI. Without knowing how many images would show up, participants were asked to judge if the last image shown changed from the previous image or not. Critically, the second-to-last image on each trial (here, the fifth frame) could be the pre-boundary, boundary, or post-boundary frame of the action. (B) When a change occurred at the boundary frame, participants were less likely to identify it, similar to Experiment 2. (C) Mean accuracy reduced by 16.7% at boundary across participants. (D) All participants, except one, were less sensitive to changes of boundary frames compared to non-boundary frames. The boundary effect was calculated as the difference in accuracy between non-boundary and boundary conditions. Error bars = 1 s.e.m.

or J on their keyboard to indicate whether the last image had shifted at all relative to the previous one. Two factors were fully crossed within-subject: 3 (pre-, on-, post-boundary) $\times$ 2 (shift, no shift) $\times$ 20 (actions) + 4 practice trials = 124 trials. Trial order was randomized across participants (except for practice trials).

4.2. Results and discussion

One participant was excluded for low accuracy ($< 60\%$, not significantly above chance), leaving 19 participants with analyzable data, with a mean accuracy of 73.7%. Across 20 actions, observers' detection accuracy at pre-boundary, boundary, and post-boundary frames was 70.8%, 57.6%, and 77.9%, respectively (4B). Disparate performance between boundary and non-boundary frames was confirmed by a paired t-test, $t(18) = 7.41$, $p = 7.14 \times 10^{-7}$, $d = 1.70$, 95% $CI_{effect} = .17[.12, .21]$ (Figure 4C). All participants, except one, showed lower sensitivity to image changes at boundary compared to non-boundary frames (Figure 4D).

A repeated-measures ANOVA was conducted to assess how accuracy varied across three temporal conditions: on-boundary, post-boundary, and pre-boundary. This exploratory analysis revealed a significant main effect of condition, $F(2, 36) = 28.17$, $p < .001$, $partial\eta^2 = .61$. Post-hoc comparisons indicated that accuracy was significantly lower during the on-boundary condition compared to the post-boundary (Tukey-adjusted $p < .0001$) and pre-boundary conditions ($p = .0006$). The difference between pre and post conditions was marginal and did not reach statistical significance ($p = .058$).

Though we do not read too deeply into this latter result as the difference was marginal, the higher accuracy right after the action boundary (relative to pre-boundary) is consistent with the predictions of Event Segmentation Theory, where a transient increase in prediction error at event boundaries opens a perceptual gate, allowing the current event representation to be updated (Reynolds et al., 2007; Zacks, 2020). This effect might have been concealed in Experiment 2 where observers viewed the full action and then made offline judgments afterward. Here, observers viewed image sequences up to the target frame, and thus their increased visual sensitivity to image features at the post-boundary frames may reflect how the action boundary functions to update an internal event model.

Overall, the results suggest that the perception of atomic action boundaries interrupted the processing of subtle image differences, even when viewing static stimuli. Unlike Experiment 2, observers made their responses here in the absence of any image motion signals, and were never given a chance to preview the complete action videos. Yet, when the feature change happened around the boundary frame, they were less likely to detect it.

353 5. Experiment 4a: Low-level dynamic features

354 Experiments 2-3 revealed a perceptual effect wherein participants were less likely
355 to detect subtle changes at action boundaries compared to non-boundaries. However,
356 are such effects merely driven by salient visual differences between boundary and
357 non-boundary frames? If frame-to-frame changes are more subtle around action
358 boundaries (i.e., slower movement of pixels at boundaries), transient slowdowns at
359 boundary frames would be less noticeable than those at non-boundary frames. In
360 this experiment, we introduced a method to generate “spiralized” videos, whereby
361 pixel-level dynamics are sufficiently preserved while high-level semantic information
362 is removed from the stimuli, allowing us to ask whether boundary effects were fully
363 explained by visual cues in the stimuli or not.

364 5.1. Method

365 5.1.1. Participants

366 In line with our pre-registration, 35 participants were recruited through Prolific.
367 A power analysis on the main results of a smaller pilot (i.e., the pre-registered paired
368 t-tests on the boundary effect) suggested that this sample would have power above
369 95% to reveal differences between normal and spiralized videos in terms of boundary
370 effects (considering accidental loss of data). For Experiments 4a-b and 5b, which
371 used the same number of conditions and trials, we pre-registered the same sample
372 size.

373 5.1.2. Stimuli and procedure

374 We used the same videos as earlier experiments but distorted all the frames of
375 the actions with a spiral-shaped filter, generating a new set of videos that preserved
376 most lower-level motion signals in the original videos while removing higher-level
377 semantic information. The spiralized videos were created by taking each frame of
378 the given standard action video clip and running it through a twirl filter with 600°
379 using Adobe Photoshop (Figure 5A).

380 Indeed, the distorted videos maintained a nearly identical frame-to-frame pro-
381 file in terms of the pixel changes across pairs of frames as the original actions.
382 For example, in the golf swing stimulus, the obvious change in pixel movements
383 between the backswing and the downswing is shown as the transition from clear
384 outward to inward motion in its spiralized version, and the deceleration and ac-
385 celeration during the swing are similarly salient in the spiral pattern. In fact,
386 pixel changes across consecutive frames were highly correlated between original and
387 spiralized videos ($r = 0.97$ across 20 actions), giving rise to highly similar im-
388 pressions of motion and dynamics. In some cases, the motion signal was even

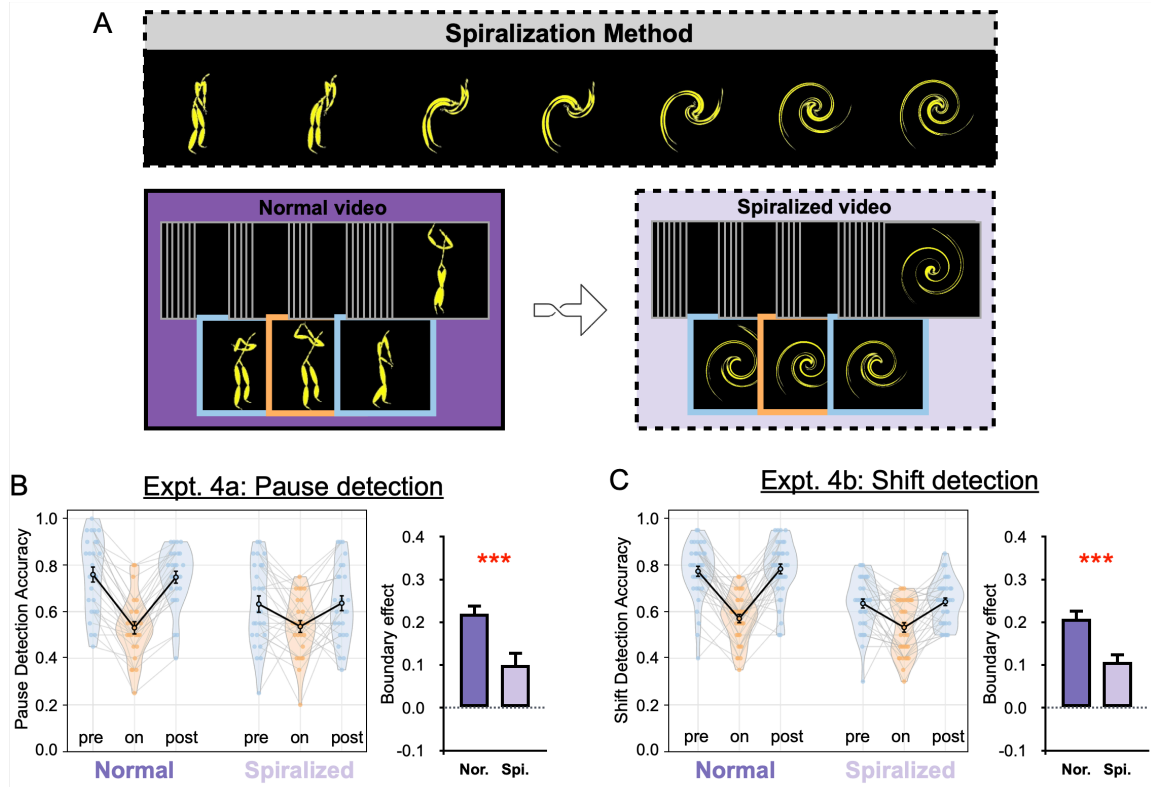


Figure 5: (A) Method used to generate the spiralized video stimuli. For each action video, we radially twisted all its frames using a spiral-shaped filter, rendering an abstract, meaningless pattern that maintained many of the lower-level spatiotemporal dynamics of the video clips. Experiments 4a and 4b used the same design as Experiment 2 and 3 respectively, except that observers saw the spiralized versions of videos and images. (B) In Experiment 4a, the boundary effect (i.e., the reduced detectability of pauses at boundaries relative to non-boundaries) was significantly weakened in the altered videos. (C) In Experiment 4b, the boundary effect decreased for the spiralized images. The boundary effect was calculated as the difference in accuracy between non-boundary and boundary conditions. Error bars = 1 s.e.m. Asterisks indicate significant differences between means ($***p < .001$).

stronger in the spiralized videos due to the stretching of the pixels (on average, the between-frame shift is 3824 pixels in normal videos versus 4350 pixels in spiralized videos). Readers can experience the dynamics in normal and spiralized videos at: <https://zk.actlabresearch.org/segmentation/skeleton.html>.

We quantified pixel-level motion features at the pre-boundary, boundary, and post-boundary frames by calculating the number of pixels that changed relative to the preceding (-1) and following ($+1$) frames. Across the 20 actions, pixel change was numerically greater at the pre- and post-boundary frames than at the boundary frame. For normal videos, the average number of changed pixels was 3907 (pre-boundary), 3701 (boundary), and 3921 (post-boundary). For spiralized videos, the values were 4435 (pre-boundary), 4206 (boundary), and 4462 (post-boundary). However, the pixel-change difference between boundary and non-boundary frames were not statistically significant across 20 videos (all P s > 0.1). The unique contribution of pixel-level features were further examined in an exploratory regression analysis reported below.

Observers attempted to detect transient pauses in the second play of each video. The experimental design was the same as Experiment 2, except that: 1) two types of videos were used — normal videos versus spiralized videos, and 2) for each participant, 10 randomly-selected actions used 60 ms for the pause trials and the other 10 used 90 ms. The Video Type condition was crossed with the other conditions as in the previous experiment, yielding $20 \text{ (actions)} \times 3 \text{ (pre-, on-, post-boundary)} \times 2 \text{ (with/without pause)} \times 2 \text{ (normal, spiralized video)} + 4 \text{ practice trials} = 244 \text{ trials}$. We imposed brief pauses either at the pre-boundary, boundary, or post-boundary frames in both the normal and spiralized video conditions, using the same frame locations for each condition. Trial order was randomized across participants (except for practice trials).

5.2. Results

Eight participants were excluded for low accuracy ($< 57\%$, not significantly above chance according to a binomial test with $\alpha = .05$) and 1 for failing to provide complete data, leaving 26 participants with analyzable data, with a mean accuracy of 75.3%. Significant boundary effects were found in both types of videos: Observers were less accurate in detecting brief pauses at boundary frames relative to non-boundary frames (normal video: $t(25) = 11.05$, $p = 4.12 \times 10^{-11}$, $d = 2.17$, 95% $CI_{effect} = .22[.18, .26]$; spiralized video: $t(25) = 3.16$, $p = .004$, $d = 0.62$, 95% $CI_{effect} = .10[.041, .16]$). However, the boundary effect (i.e., the decreased detection accuracy at boundary relative to non-boundary frames) was significantly weakened for the

425 spiralized stimuli ($t(25) = 4.19$, $p = .00030$, $d = 0.82$, $p = .004$, $d = .62$, 95%
 426 $CI_{difference} = .12[.059, .18]$)(Figure 5B).

427 Two follow-up exploratory analyses were used to further examine the data. First,
 428 we conducted a 2 (normal, spiralized) $\times$ 3 (pre-boundary, boundary, post-boundary)
 429 repeated-measures analysis of variance (ANOVA), which revealed significant main
 430 effects of Testing Frame, $F(2, 50) = 35.21$, $p < .001$, $\eta_p^2 = .58$, and Video Type,
 431 $F(1, 25) = 18.05$, $p < .001$, $\eta_p^2 = .42$, as well as a significant Testing Frame $\times$ Video
 432 Type interaction, $F(2, 50) = 11.52$, $p < .001$, $\eta_p^2 = .32$. Follow-up pairwise com-
 433 parisons showed that in the normal condition, accuracy was significantly lower at
 434 the boundary compared to both the pre-boundary (Tukey-corrected $p < .0001$) and
 435 post-boundary time points ($p < .0001$). In the spiralized condition, accuracy at the
 436 boundary was numerically lower than both pre- and post-boundary, but these com-
 437 parisons did not reach significance thresholds ($p = .082$ and $p = .053$, respectively).
 438 Finally, accuracy was significantly higher in the normal condition than the spiralized
 439 condition at both the pre-boundary ($p = .0015$) and post-boundary ($p = .0006$)
 440 frames (Figure 5B).

441 Second, a multiple linear regression model was used to examine the extent to
 442 which Video Type (normal vs. spiralized, coded as 1 and 0), Testing Frame (pre-,
 443 on-, post-boundary, coded as 1, -1, 1) and Pixel Change (see 5.1.2 for details about
 444 how this metric is computed) explained variance in response accuracy. The overall
 445 model explained approximately 24.7% of the variance in accuracy ($R^2 = 0.247$).
 446 All three predictors were statistically significant. Accuracy was significantly lower
 447 for twisted videos compared to normal ones ($\beta = -0.13$, $p < .001$), and higher at
 448 non-boundary frames ($\beta = 0.08$, $p < .001$) compared to boundaries. Pixel change
 449 also showed a modest but significantly positive relationship with response accuracy
 450 ($\beta = 0.00006$, $p = .022$). Relative importance analysis revealed that Testing Frame
 451 uniquely accounted for the largest proportion of explained variance (58.3%), followed
 452 by Video Type (29.6%) and Pixel Change (12.1%).

453 These results showed that action boundaries impaired performance more reliably
 454 in normal videos compared to spiralized videos where semantic content was removed.
 455 This suggests that the perception of event boundaries in single brief actions may not
 456 solely rely on pixel-level, spatiotemporal changes in stimuli, but also on internal rep-
 457 resentations of event structure; in this case, the structure of familiar bodily actions.

458 6. Experiment 4b: Low-level static features

459 To control for pixel-level change of visual features involved in the image shift
 460 paradigm, we again used spiralized images in a task similar to Experiment 3, al-

lowing us to compare the boundary effect in normal images versus their spiralized counterparts.

6.1. Method

6.1.1. Participants

As stated our pre-registration, 35 participants were recruited through Prolific.

6.1.2. Stimuli and procedure

The experimental design was the same as Experiment 3, except that two types of images were used: normal versus spiralized. We quantified pixel-level change at the pre-boundary, boundary, and post-boundary frames by calculating the number of changed pixels between the last two images in the sequence (2 frames apart in the original video). Across the 20 actions, pixel change was numerically greater at the pre- and post-boundary frames compared to the boundary frame. For normal videos, the average number of changed pixels was 4231 (pre-boundary), 4018 (boundary), and 4188 (post-boundary). For spiralized videos, the corresponding values were 4816 (pre-boundary), 4570 (boundary), and 4778 (post-boundary). However, the pixel-change difference between boundary and non-boundary frames were not statistically significant (all P s ≥ 0.1). The unique contribution of pixel-level features were further examined in the exploratory analysis, as reported below.

The Image Type condition was crossed with other conditions as in previous experiments, yielding 20 (actions) $\times 3$ (pre-, on-, post-boundary) $\times 2$ (shift, no shift) $\times 2$ (normal, spiralized image) + 4 practice trials = 244 trials. Participants were instructed to detect whether the last image in the sequence was identical to the one immediately preceding it (Figure 4A). Trial order was randomized across participants (except for practice trials).

6.2. Results and discussion

One participant was excluded for failing to submit complete data, leaving 34 participants with analyzable data, with a mean accuracy of 72.4%. Thirty-four observers attempted to detect image changes across both conditions.

We again observed a significant boundary effect in the normal images ($t(33) = 10.31$, $p = 7.52 \times 10^{-12}$, $d = 1.77$, 95% $CI_{effects} = .21[.17, .25]$), and also observed this effect in the spiralized images ($t(33) = 5.04$, $p = 1.63 \times 10^{-5}$, $d = 0.86$, 95% $CI_{effects} = .11[.065, .15]$); however, the effect was about half as strong in the spiralized images versus the normal images ($t(33) = 3.59$, $p = .0011$, $d = 0.62$, 95% $CI_{difference} = .10[.045, .16]$, Figure 5C), echoing the results of Experiment 4a.

495 We then conducted two follow-up exploratory analyses. First, a 2 (normal, spiral-
496 ized) $\times$ 3 (pre-boundary, boundary, post-boundary) repeated-measures ANOVA re-
497 vealed significant main effects of Testing Frame, $F(2, 66) = 50.29$, $p < .001$, $\eta_p^2 = .60$,
498 and Image Type, $F(1, 33) = 40.30$, $p < .001$, $\eta_p^2 = .55$. Importantly, there was also
499 a significant Testing Frame $\times$ Image Type interaction, $F(2, 66) = 7.73$, $p < .001$,
500 $\eta_p^2 = .19$, indicating that the effect of frame position varied across image types.
501 Post hoc comparisons showed that in the normal condition, accuracy was signif-
502 icantly lower at the boundary compared to both pre-boundary ($p < .0001$) and
503 post-boundary ($p < .0001$) frames. Similarly, in the spiralized condition, boundary
504 accuracy was significantly lower than both pre-boundary ($p = .0062$) and post-
505 boundary ($p = .0002$), but the differences were smaller in magnitude. As can be seen
506 in Figure 5C, observers were significantly more accurate in the normal condition than
507 the spiralized condition at both the pre-boundary ($p < .0001$) and post-boundary
508 ($p < .0001$).

509 Second, a multiple linear regression was conducted to examine the extent to which
510 Image Type, Testing Frame, and Pixel Change accounted for variance in response
511 accuracy. The overall model was significant, $F(3, 116) = 8.14$, $p < .001$, $R^2 = 0.174$.
512 All three predictors made significant contributions: accuracy was lower for twisted
513 videos compared to normal ones ($\beta = -0.11$, $p = .010$), higher at non-boundary
514 frames ($\beta = 0.073$, $p = .001$) compared to boundaries, and increased slightly with
515 greater pixel change ($\beta = 0.00005$, $p = .041$). Relative importance analysis indicated
516 that frame position accounted for the largest share of explained variance (50.5%),
517 followed by video type (27.8%) and pixel change (21.7%). The results revealed a
518 robust unique contribution of image type to the boundary effect.

519 Boundary effects were present under both conditions but were more pronounced
520 in normal action videos. The difference of boundary effects between normal versus
521 spiralized stimuli mostly came from observers' lower accuracy in detecting changes
522 of spiralized non-boundary frames (Figure 5), suggesting that observers did better
523 predicting the flow of actions when semantic information was present, and thus were
524 more sensitive to subtle changes within action steps. Taken together, these results
525 suggest that in addition to pixel-level changes and motion cues, mental representa-
526 tions of internal structures in atomic actions may be involved in rapid, spontaneous
527 visual segmentation.

528 7. Experiment 5a: Point-light displays

529 Our last two experiments aimed to replicate and extend our findings to the case of
530 point-light displays. Although the twisting procedure used in Experiment 4a and 4b

maintained large-scale pixel-level and spatiotemporal changes in the action videos, it inevitably altered the percept of biological motion and other visual features, like motion direction and figure rigidity. In this experiment, we re-created the 20 actions as “point-light walkers” (PLWs) (Johansson, 1973; Van Boxtel & Lu, 2013) and attempted to replicate the previous effects. Each action video now consisted of a number of coordinated moving points that represent the joints of the human figures. In this experiment, we presented PLWs either upright or upside-down. Inverted PLWs serve as an interesting case wherein the semantic information of actions is only partially preserved and can impede recognition (Shipley, 2003; Sumi, 1984). We asked whether the boundary effect emerged in visually minimal PLW stimuli, and whether the effect differed between upright versus inverted stimuli.

7.1. Method

7.1.1. Participants

Consistent with our pre-registration, 80 participants were recruited through Prolific. We increased the sample size for this experiment based on the preliminary result of a small pilot study.

7.1.2. Stimuli and Procedure

This experiment transformed the 20 original action videos into “point-light walker” (PLW) stimuli. This was implemented using the BioMotion MATLAB toolbox (Van Boxtel & Lu, 2013). The number of joints used to define actions ranged from 20 to 30. This new set of videos depicted the same 20 actions as the ones used in previous experiments, yet were slightly different in the number of frames as a higher sampling rate was used. The boundary frame was hand-selected as the frame that had the matching posture as the boundary frame used in previous experiments, and the pre- and post-boundary frames were selected as before.

The PLW stimuli were presented either upright or upside-down. To reduce the number of repetitions of each action video (which could lead to recognition of the inverted stimuli), we asked participants to view each action only once in each trial and detect if the video was briefly frozen at any point. A 120-ms pause was applied to pre-, on-, and post-boundary frames of each action video. All 20 actions appeared for each of these trial types, for $20 \text{ (actions)} \times 3 \text{ (pre-, on-, post-boundary)} \times 2 \text{ (with, without pause)} \times 2 \text{ (upright, inverted video)} + 4 \text{ practice trials} = 244 \text{ trials}$. Crucially, we note that because the experiments were crowd-sourced, we could not monitor participants for any behavior that would change the perceived orientation of the video clips (e.g., head-tilting).

7.2. Results and discussion

18 participants were excluded for low accuracy ($< 57\%$) and 1 participant for failing to submit a complete data set, leaving 61 participants with analyzable data, with a mean accuracy of 66.1%. The boundary effects arose similarly in both the upright videos ($t(60) = 13.34, p = 1.81 \times 10^{-19}, d = 1.72, 95\% CI_{effects} = .24[.21, .28]$) and inverted videos ($t(60) = 11.35, p = 1.82 \times 10^{-16}, d = 1.47, 95\% CI_{effects} = .21[.17, .25]$). As predicted, the boundary effect was numerically stronger in upright videos versus inverted videos (24.4% versus 21.1%), however, the difference was only marginally reliable given our pre-registered criteria ($t(60) = 1.75, p = .085, d = .22, 95\% CI_{difference} = .033[-.004, .07]$). An exploratory analysis using Bayesian paired-sample t-test showed weak evidence for the null hypothesis; ($BF_{10} = .59$, using JZS prior with scale $\sqrt{2}/2$).

A 2 (upright, inverted) $\times$ 3 (pre-boundary, boundary, post-boundary) repeated-measures ANOVA revealed a robust main effect of Testing Frame, $F(2, 120) = 120.44, p < .001, \eta_p^2 = .67$. There was no main effect of Video Type, $F(1, 60) = 0.12, p = .73, \eta_p^2 = .002$. However, there was a significant Testing Frame $\times$ Video Type interaction, $F(2, 120) = 3.82, p = .025, \eta_p^2 = .06$, suggesting that the effect of condition was indeed modulated by orientation. Post-hoc comparisons revealed that in the upright condition, accuracy at the boundary was significantly lower than both the pre-boundary ($p < .0001$) and post-boundary frames ($p < .0001$). In the inverted condition, a similar pattern was observed: accuracy at the boundary was significantly lower than both pre-boundary ($p < .0001$) and post-boundary ($p < .0001$). This result implies that spatiotemporal information significantly contributes to perceptual segmentation in this set of stimuli. It may be that partially preserved semantic information in inverted actions — particularly when it is potentially recovered through repeated exposure to the stimuli — as well as the lack of control for head tilting movements may help explain why the boundary effect remained comparably robust in the inverted condition.

8. Experiment 5b: Scrambled PLWs

Upside-down PLW actions are less recognizable yet still preserve partial semantic information (e.g., people may see still actions but misperceive some properties; See Barclay et al., 1978; Sumi, 1984). Is there a better way to eliminate semantic information in PLW stimuli that avoids these limitations? Here, we shuffled the initial positions of the “joints” such that the configuration of the body shape no longer resembled the human figure, while the local joint motion and rigidity was maintained.

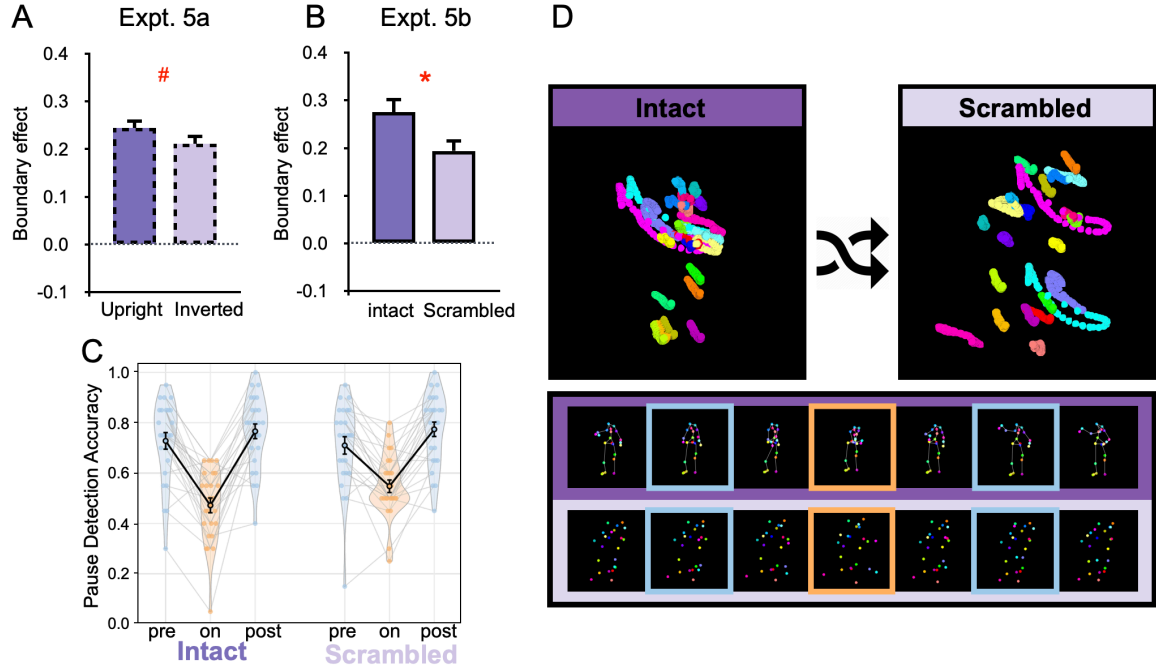


Figure 6: (A) Results of Experiment 5a. The reduction of boundary effect in upside-down point-light walker stimuli was marginally significant relative to upright stimuli. (B&C) Results of Experiment 5b. The reduction of accuracy occurred at boundary for both intact and scrambled videos, but the size of reduction differed between two conditions. The boundary effect was weaker when the global configuration of the observed agent was disrupted. (D) Example stimuli used in Experiment 5b. For a given action, the movements of all the individual points (joints) were identical between its intact form and scrambled form. Here, each color represents a unique joint, and the exact movement trajectory of each colored joint across all frames was identical between two types of stimuli. Similar to Experiment 2, a brief pause could occur at the pre-boundary, boundary, or post-boundary frame. (Note that all the visual points of the actual stimuli were colored yellow as in the original skeletonized stimuli, and the underlying skeleton, indicated by the thin lines in the above figure, was not visible to observers.) The boundary effect was calculated as the difference in accuracy between non-boundary and boundary conditions. Error bars = 1 s.e.m. Asterisks indicate significant differences between means for two-tailed t-test ($*p < .05$; $\#p < .1$)

This method fully removes the semantic content related to human actions while preserving the unique motion patterns of the biological movements (producing a bizarre action made by some kind of “creatures”, see Chen et al., 2022). In this case, observers lack internal knowledge for the event structure but can still rely on biological motion cues. Thus, from the observer’s perspective, the spatiotemporal motion perceived in the intact point-light displays is preserved in the scrambled videos, but the stimuli lose semantic meaning.

8.1. Method

8.1.1. Participants

Consistent with our pre-registration, 35 participants were recruited through Prolific.

8.1.2. Stimuli and procedure

The point-light videos used in Experiment 4a were made into “scrambled” versions, where the initial positions of the joints were randomly selected — separately by x, y and z — from the ranges of width, height, and depth of the original videos (Figure 6A). Readers can experience the intact and scrambled stimuli at <https://zk.actlabresearch.org/segmentation/plw.html>.

The experimental design was the same as Experiment 4a, except that: 1) point-light stimuli (normal and scrambled videos) were used, and 2) the pause duration was always 75 ms (to counter the overall increased difficulty of the task). Similar to Experiment 4a, all 20 actions appeared for each of these trial types, for 20 (actions) $\times$ 3 (pre-, on-, post-boundary) $\times$ 2 (with, without pause) $\times$ 2 (normal, scrambled video) + 4 practice trials = 244 trials. Trial order was randomized across participants (except for practice trials).

8.2. Results

Ten participants were excluded for low accuracy ($< 57\%$), leaving 25 participants with analyzable data, with a mean accuracy of 76.6%. We again observed robust boundary effects in both the intact ($t(24) = 9.02$, $p = 3.53 \times 10^{-9}$, $d = 1.80$, 95% $CI_{effects} = .28[.22, .33]$) and scrambled videos ($t(24) = 8.11$, $p = 2.45 \times 10^{-8}$, $d = 1.62$, 95% $CI_{effects} = .19[.15, .24]$). Critically, these boundary effects were reduced in the scrambled videos, where semantic action structure was not perceivable ($t(24) = 2.22$, $p = .036$, $d = 0.44$, 95% $CI_{difference} = .081[.01, .15]$, Figure 6C).

633 We also conducted a 2 (intact, scrambled) $\times$ 3 (pre-boundary, boundary, post-
634 boundary) repeated-measures ANOVA, as a secondary analysis². This revealed a
635 strong main effect of Testing Frame, $F(2, 48) = 55.20$, $p < .001$, $\eta_p^2 = .70$, and
636 a marginal Testing Frame $\times$ Video Type interaction, $F(2, 48) = 2.83$, $p = .069$,
637 $\eta_p^2 = .11$. The main effect of video type was not statistically significant, $F(1, 24) =$
638 2.46 , $p = .13$, $\eta_p^2 = 0.09$. Follow-up comparisons revealed that, for intact videos,
639 accuracy at the boundary was significantly lower than both the pre- ($p < .0001$)
640 and post-boundary moments ($p < .0001$). For scrambled videos, accuracy at the
641 boundary was also significantly lower than both the pre- ($p = .0001$) and post-
642 boundary ($p < .0001$). The numerically larger boundary drop for intact videos
643 (e.g., accuracy at boundary between intact and scrambled conditions, $p = .085$) is
644 consistent with the observed interaction effect (Figure. 6B).

645 Though biological motion is thought to be processed as a special category (Blake-
646 more & Decety, 2001), it cannot easily explain the stronger boundary effect observed
647 in the real actions relative to the scrambled biological movements. Thus, while we
648 observed a significant contribution of spatiotemporal dynamics and biological motion
649 to the visual segmentation (i.e., significant boundary effects in the scrambled condi-
650 tion), our results demonstrate that semantic knowledge may also play a significant
651 role in the perception of action event structure.

652 9. Discussion

653 In this work, we asked if the event structure within single, brief “atomic” ac-
654 tions is represented in visual perception, and if this representation involves semantic
655 knowledge. Using motion-captured video stimuli, static action images, and biologi-
656 cal motion stimuli, we consistently observed that the transition of an action’s salient
657 internal sub-routines impaired observers’ ability to visually detect subtle changes in
658 the stimuli, suggesting that the event structure within these brief actions is indeed
659 represented in the visual system. Control studies suggested that this boundary rep-
660 resentation is not solely driven by basic visual features of the stimulus that happen
661 to change at the boundary (e.g., spatiotemporal features and motion cues), but also
662 by internal models of the actions themselves (Experiments 4a–b and 5a–b). That

²In addition to the main analysis, we reported repeated-measures ANOVA across participant means for detection accuracy for Experiment 4a-b and 5a-b. Note that we determined sample size only using power analyses on the main results (i.e., the pre-registered paired t-tests on the boundary effect) using pilot studies. Thus, the exploratory ANOVAs in some studies likely do not have sufficient power with the current sample size.

is, semantic information of human actions, even very brief actions, plays a key role in driving perceptual effects at event boundaries, and thus may support automatic segmentation in visual processing.

9.1. Breaking down “minimal events”

In their seminal work, Zacks and Tversky (2001) raised two fundamental questions about event perception: (a) What are the basic units of actions? And (b) What are events? They proposed that the atomic components of events are *basic actions*, such as raising one’s hand or moving one’s head. Later empirical studies supported this view, showing that atomic actions tend to be single actions with well-defined starts and ends, similar to those used here. When observers segment streams of behavior into coarse and finer segments, fine-grained boundaries often arise between single actions like “walks into the room”, “open door”, “take out food”, etc. as in Zacks et al. (2001). More recently, it has been proposed that bounded events (e.g., “scoop up yogurt”, “fold a handkerchief”, “roll up a towel”) constitute the basic units or “mental individuals” in the temporal domain (Lee et al., 2024, see a related work about “objecthood” of auditory events in Goh et al., in press). This idea, that events can be considered objects in time, dates back to earlier philosophical analyses (Quine, 1985).

In contrast to a large body of literature that has explored event segmentation *between* distinct actions (D. Baldwin et al., 2008; Buchsbaum et al., 2015; Franklin et al., 2020; Lea et al., 2016; Newton, 1973; Newton et al., 1977; Pomp et al., 2024; Swallow et al., 2009; Wang et al., 2013; Zacks et al., 2009), the present work broke down the minimal units in event perception and focused on event boundaries that occur *within* short, bounded (atomic) actions (Y. Ji & Papafragou, 2022; Vendler, 1957). Through this approach, we demonstrate that atomic actions are not indivisible events; but instead, they are represented as having their own internal event structure. Consistent with prior work that conceptualized objects and events as individuated regions in space and time, respectively, our results might point to an even deeper parallel between objects and events: As the visual system represents a compositional spatial structure within single objects (Biederman, 1987; Feldman & Singh, 2006; Sun & Firestone, 2021), it may also represent components and internal structure within single actions in the time domain.

Our finding arguably fits with the recent resurgence of interest in ideas like the Language-of-Thought approach to vision, which proposes multiple types of compositional and structured representations within perception (Hafri et al., 2024; Papeo et al., 2024; Quilty-Dunn et al., 2023). While most evidence for LoT-like representation in perception has focused on the spatial domain (Hafri & Firestone, 2021;

700 Lovett & Franconeri, 2017; Strickland & Scholl, 2015; Sun et al., 2025), the current
 701 study presents preliminary evidence that visual perception spontaneously represents
 702 basic episodes of brief experiences as discrete, inherently structured events unfolding
 703 over time. By extending the idea that visual perception possesses potential LoT-like
 704 properties to fundamental temporal units, this work points to a broader presence of
 705 compositional, structured representations across perception.

706 **9.2. *Perceptual effects in event segmentation***

707 Our study builds on a number of studies that have discussed the spontaneous
 708 nature of representing event boundaries in perception. A number of experiments
 709 have shown that the representation of the transition between events interferes with
 710 the perceptual detection of subtle disruptions, resulting in a lower detection accuracy
 711 in detecting disruptions at boundary versus non-boundary time-points within events,
 712 seen in both visual perception (Huff et al., 2012; Y. Ji & Papafragou, 2022; Yates
 713 et al., 2024) and auditory perception (Repp, 1992, 1998). For example, observers are
 714 less sensitive to visual interruptions at the end points of events (e.g. a girl folding her
 715 handkerchief) compared to the middle points (Y. Ji & Papafragou, 2022) (but also
 716 see Hard et al., 2019; Ongchoco et al., 2023; Swallow et al., 2009 and Footnote #2
 717 for middle frame effects that may interact with boundary effects). This perceptual
 718 effect also is found in physical events: The boundary of simple physical events (such
 719 as collision, containment, or falling) has been found to impede the processing of
 720 distracting information (e.g., unnatural pauses at boundary timepoints) (Yates et
 721 al., 2022). Such results have been explained by the rise of prediction error at the
 722 boundary and subsequent enhanced attention to event information (Antony et al.,
 723 2021; Pradhan & Kumar, 2022; Reynolds et al., 2007; Zacks et al., 2007, 2011).

724 However, in all of the above studies, feature, motion, and movement informa-
 725 tion was inevitably correlated with the event structure in ongoing visual input. Our
 726 study thus contributes to and builds on this line of work by attempting to dissoci-
 727 ate two possible mechanisms that may account for the perceptual consequences of
 728 event segmentation: spatiotemporal dynamics and motion cues within the stimulus,
 729 and internal representations of semantic event structure. Our findings suggest that
 730 predictive processes that define discrete events during visual perception may employ
 731 both visual features and semantic content to generate predictions.

732 While our study built on the logic of previous studies which focused on the per-
 733 ception of disruptions or distractions at event boundaries (Huff et al., 2012; Repp,
 734 1992; Yates et al., 2024), event structure and boundary can influence perception in
 735 other ways. For example, an early study found that observers were more accurate at
 736 detecting frame deletion when the missing frames occurred at “breakpoints” in com-

plex action sequences (e.g., a person setting out tools), compared to non-breakpoints (Newton, 1973). Enhanced perceptual encoding at an event boundary has also been shown through improved recognition performance in later memory tasks (Swallow et al., 2009). The interpretation of these effects as reflections of attentional or predictive processes could be better fleshed out; future work could look for other perceptual effects at boundaries in observed actions, such as increased processing of certain features versus others (Baker & Levin, 2015; Yates et al., 2024), and the creation of temporal distortions (Goh et al., [in press](#); Ongchoco et al., 2023a).

9.3. *Semantic structure in action perception*

Though semantic knowledge is the key source of information underlying event cognition (e.g., “Sally was sitting, now she is standing”), it was not a foregone conclusion that the automatic perceptual boundaries in our atomic action stimuli would also be shaped by semantics. First, the action segmentation in our study was even more rapid and finer-grained than those described as “fine” units in previous work (Hard et al., 2011; Yates et al., 2024; Zacks & Tversky, 2001; Zacks et al., 2009). Fine boundaries are thought to be more perceptually determined and driven primarily by external changes of pixel-level movement and motion (Hard et al., 2006; Papeo et al., 2024; Zacks et al., 2009). Second, our work used simple, abstract animations of actions in a stripped-down context, rather than the complex, semantically-rich scenes in naturalistic stimuli. Prior studies that used more abstract, simple stimuli like ours in segmentation tasks (e.g., videos of moving geometric shapes) have revealed a stronger relation between boundary judgments and visual features within the stimuli (such as movement and motion features and dynamics), accompanied by increased activity in motion-related brain regions at event boundaries (Hard et al., 2006; Pomp et al., 2024; Zacks et al., 2006, 2009). Indeed, our results using multiple forms of control stimuli consistently pointed to rather robust effects of pixel-level and motion features. However, we also found that action schema appeared to contribute to visual event segmentation, suggesting that semantic information plays a prominent role in the automatic visual processing of action event structures.

Our work also sheds new light on our understanding of the mental representation of motor action structure. Some actions in our study exhibited a “preparation” stage followed by an “execution” phase, for example, the “focus” and “shoot” phases in a basketball shot. Empirical studies have revealed this preparation-execution structure and its function in facilitating action recognition and anticipation in ordinary actions (Cohn et al., 2017; Urgesi et al., 2010) and in athletic skills (Aglioti et al., 2008; Lasher, 1981; Smith, 2016; Urgesi et al., 2012). Yet, for most actions we examined, segmentation generally reflected discrete yet structured steps to complete an

774 action. For example, the first step of actions such as a “star jump”, a golf swing, and a
775 punch typically belongs to the execution phase, occurring after the preparation phase
776 highlighted in previous research (e.g., Lasher, 1981; Urgesi et al., 2012). Theoretical
777 work in semantic analysis has applied a more richly-structured framework of action
778 steps that may better explain our results. For instance, Jackendoff (2007) suggested
779 that simple, routine actions (e.g., shaking hands) are mentally stored in a seman-
780 tic hierarchy typically composed of three components: preparation (reaching and
781 grasping), head (shaking), and coda (releasing and withdrawing). Speculatively, our
782 findings suggest that such semantic structures exist not only as conceptual schema
783 but perhaps also within perceptual systems.

784 Our findings leave open important questions about action segmentation for future
785 research. First is the connection between segmentation and prediction in the context
786 of human actions. Given evidence that early kinematic cues can predict the outcome
787 of actions(e.g., Aglioti et al., 2008; Cohn et al., 2017), it is worth asking whether and
788 how different ways of segmenting actions might alter an agent’s ability to generate
789 accurate predictions about how an action will unfold. Related questions may include
790 whether finer segmentation ability precedes improved performance in anticipating
791 future actions, and whether skilled athletes are also more efficient in segmenting
792 others’ actions. Second, our study employed salient, simple, familiar actions; this
793 was by design, to make the semantic structure overdetermined. In future work,
794 we could ask how more novel or unfamiliar actions are segmented, or how learning
795 about new actions may affect the perception of event boundaries. Finally, how our
796 findings connect to motor learning of actions themselves could provide insights into
797 how and why people divide simple motor routines into discrete steps. For example,
798 in learning a new motor skill like a tennis serve, coaches and teachers tend to divide
799 an otherwise smooth, rapid movement into multiple discrete “chunks” based on an
800 internal structure (Fitts, 1964; Sakai et al., 2003). How do we decide where the
801 chunk boundary should be?

802 **9.4. Conclusion**

803 In sum, this work explored the perception of event structure within single atomic
804 actions: our results suggest that the mind represents rapid, continuous actions as
805 containing discrete, inherently structured steps. These structured representations
806 appear to emerge automatically over time during visual processing, and are driven
807 by both visual motion cues and the semantic structure of actions.

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CRedit authorship contribution statement

Zekun Sun: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing - Original Draft, Writing - Review & Editing, Visualization. **Samuel D. McDougale:** Conceptualization, Methodology, Investigation, Writing - Review & Editing, Supervision, Funding acquisition.

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